TASK 10 – Compare Expectations with Execution

10.1 Aggregate totals at village, taluk and district scale:

For this comparison, let us assume that each household receives an electricity bill of ₹950. This is only an example used to check the program's performance.

Scale	Number of bills	Total in paise
Village	4,000	380,000,000
Taluk	60,000	5,700,000,000
District	800,000	76,000,000,000

A normal 32-bit signed int cannot store the taluk and district totals in paise because the values are too large. A 64-bit long long can store these totals safely. A float becomes less accurate as the total increases, while a double provides better precision but cannot represent every decimal value exactly.

Before submitting, run the program on your computer and record the actual results.

Measurement	Actual result
Compiler and version	_____
sizeof(int), INT_MAX	_____
sizeof(long), LONG_MAX	_____
sizeof(long long), LLONG_MAX	_____
sizeof(float), FLT_DIG	_____
sizeof(double), DBL_DIG	_____